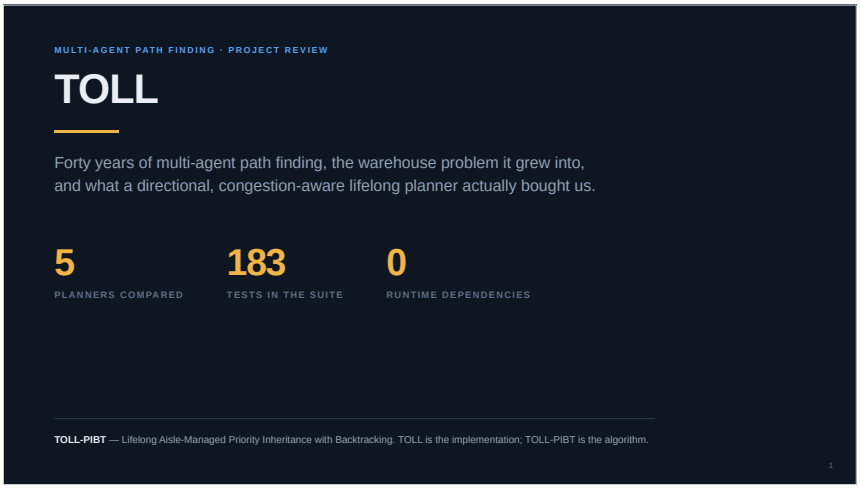
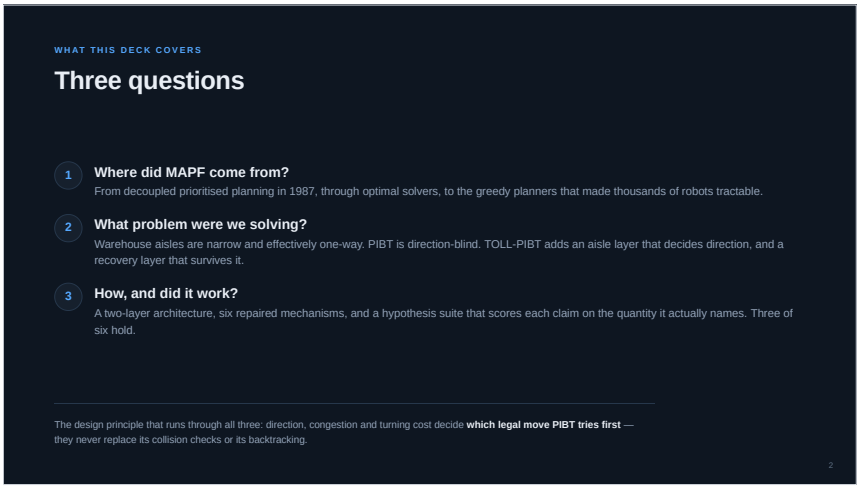


SPEAKER NOTES

Set the frame before any detail: this is a review of a research implementation, and the honest headline is that the baseline we set out to beat wins on most maps. The deck earns that admission by showing the measurements rather than burying it.

The three numbers are deliberate. Five planners because the comparison is against published work, not against ourselves. Zero dependencies because every number here can be reproduced with `python3 main.py` on a clean machine.



SPEAKER NOTES

Flag the third bullet now rather than at the end. An audience told at the start that half the hypotheses failed listens to the evidence; an audience told at slide 30 listens for what else was hidden.

THE PROBLEM

Multi-agent path finding

Given a graph and k agents, each with a start and a goal, find k paths such that every agent arrives and no two agents ever collide.

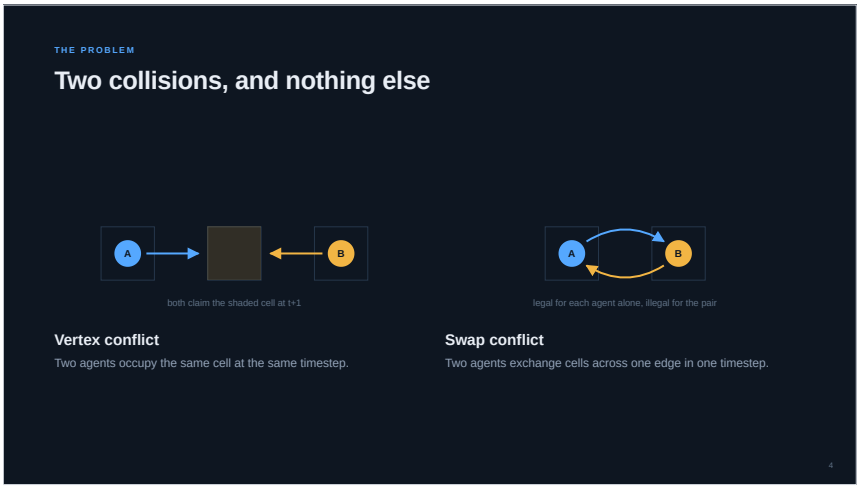
Time is discrete. At each timestep an agent moves along one edge or waits. Everything else in MAPF — makespan, sum-of-costs, or the throughput objective this project uses — is an optimisation layered on top of that one rule.

Two kinds of collision define the problem. They are the next slide, and they are the **only** two things the low level of this system checks.

3

SPEAKER NOTES

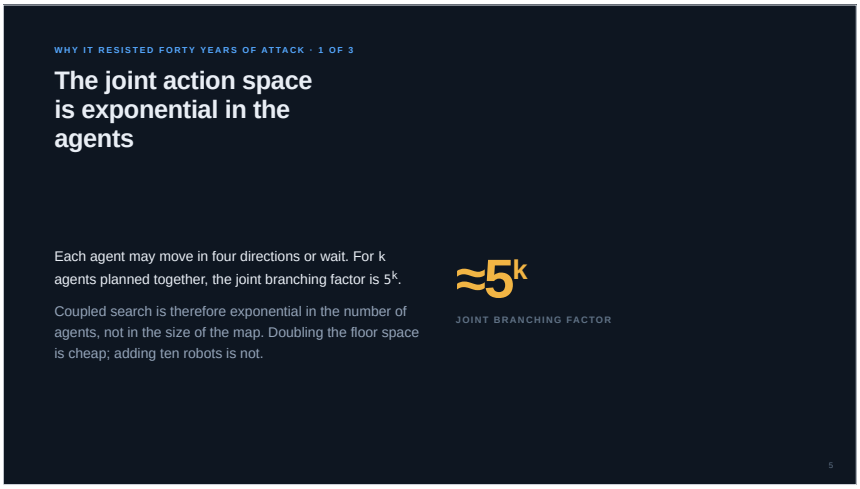
Keep this slide short on purpose. The definition is not the interesting part; the two collision types on the next slide are, because the entire architectural argument later ("the high level never decides safety") is an argument about who is allowed to enforce exactly those two rules.



SPEAKER NOTES

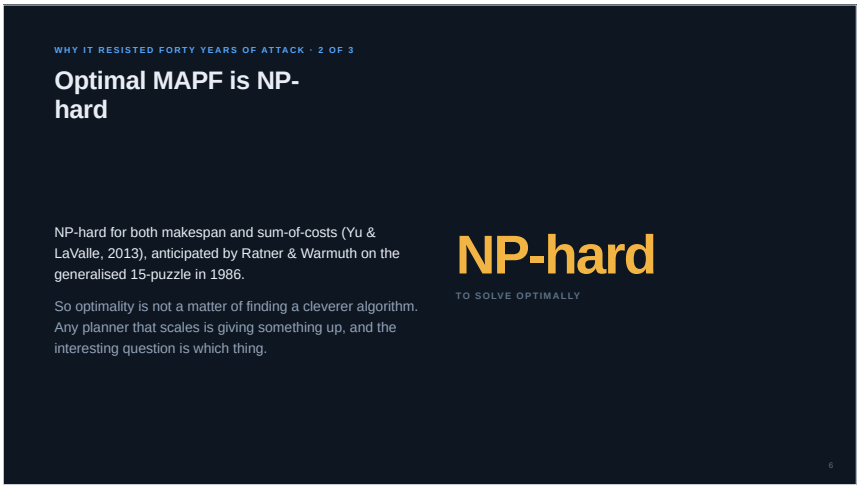
The swap conflict is the one people forget. Each agent's move is individually legal — each is moving into a cell the other is vacating — and only the pair is illegal. It is also the one that makes a naive "check the destination is empty" implementation wrong.

Worth saying out loud: these two checks are the only hard rules in our low level. Every later slide about the aisle layer is about resisting the temptation to add a third.



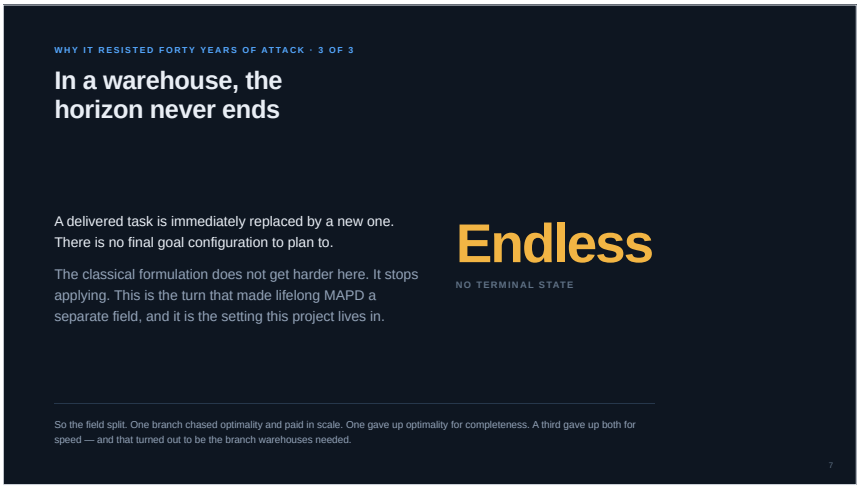
SPEAKER NOTES

Concretely: ten agents is about ten million joint actions per timestep. This is the number that pushed the field towards decoupling, and it is the number PIBT sidesteps by never planning more than one timestep ahead.



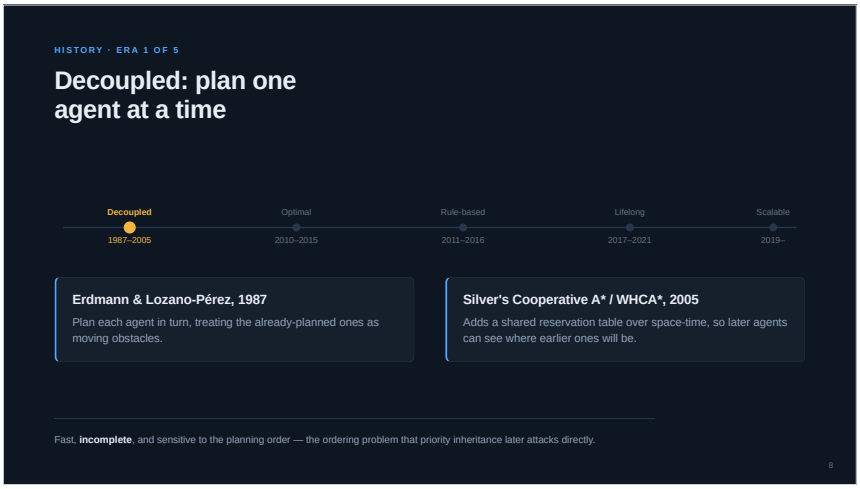
SPEAKER NOTES

Do not linger. The point is only to establish that the tradeoff table two slides on is forced, not chosen.



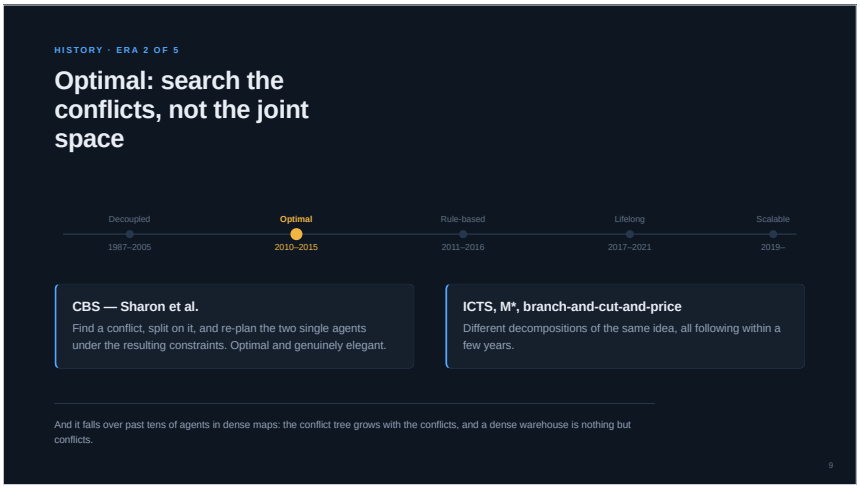
SPEAKER NOTES

This slide is the hinge of Act 1. Everything before it is classical MAPF; everything after it is about what you do when there is no plan to finish.



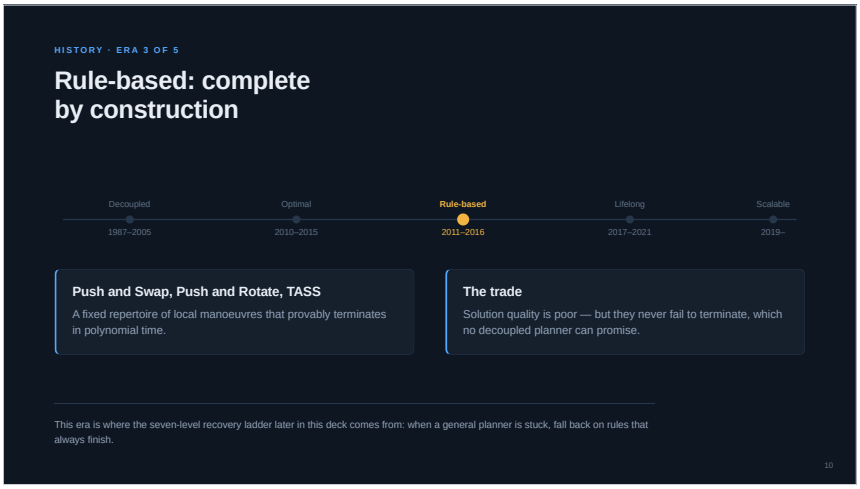
SPEAKER NOTES

The through-line to flag: this era's weakness is that the priority order is fixed in advance. PIBT's contribution, twenty years later, is letting the order change during the timestep.



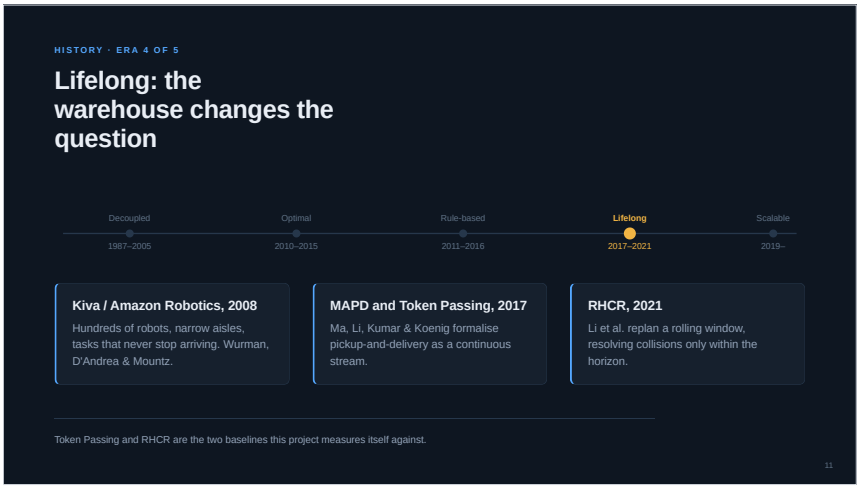
SPEAKER NOTES

If asked why we did not benchmark against CBS: at the densities in this project it does not finish. The two baselines we do run — Token Passing and RHCR — are the lifelong ones, which is the honest comparison class.



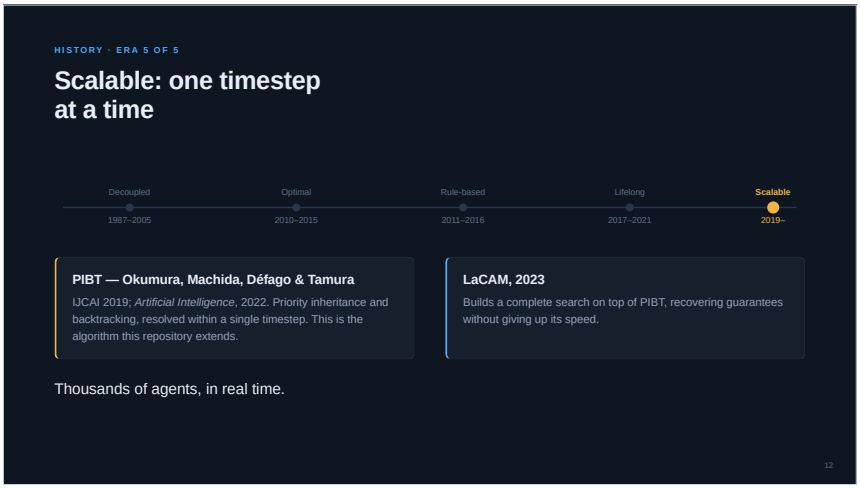
SPEAKER NOTES

Explicit link forward. Our recovery ladder is unashamedly rule-based, and it is in the system for exactly the reason this era existed.



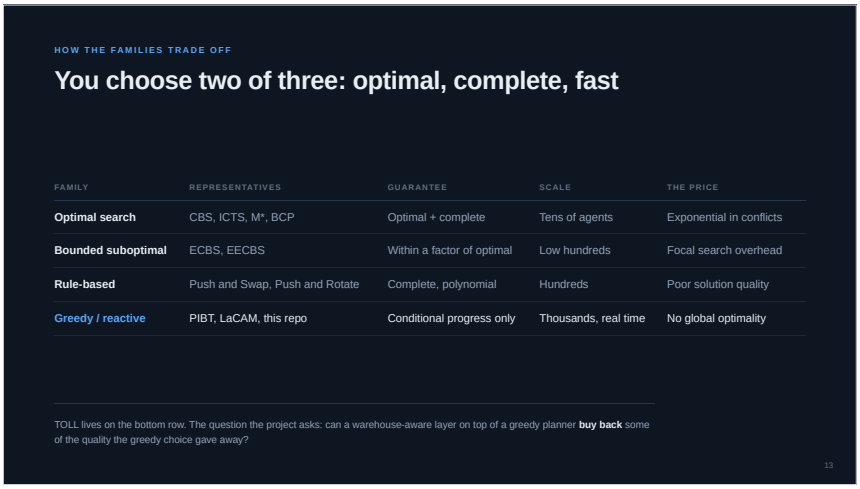
SPEAKER NOTES

Commercial pressure, not theory, produced this era. Worth saying: the reframing was forced by a real floor, and that is also why the objective changed from makespan to throughput.



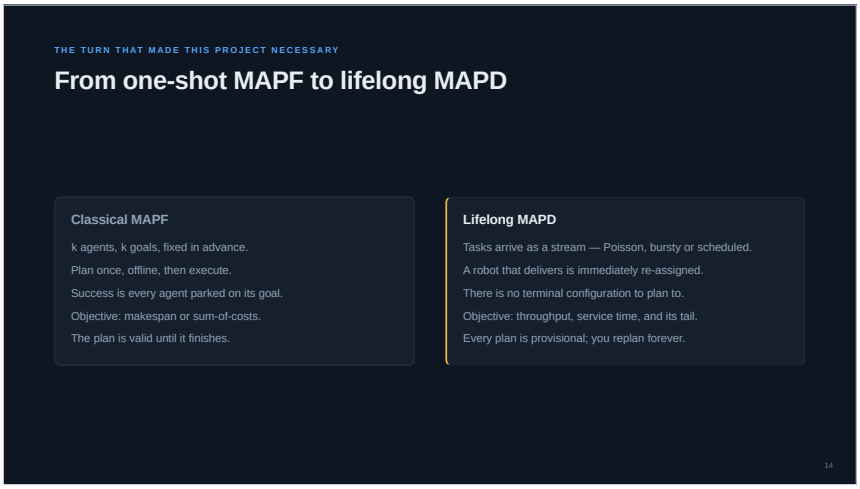
SPEAKER NOTES

PIBT is the whole basis of this project, so give it a beat. The key insight to plant here: it never plans more than one timestep, which is why the `5^k` problem never arises.



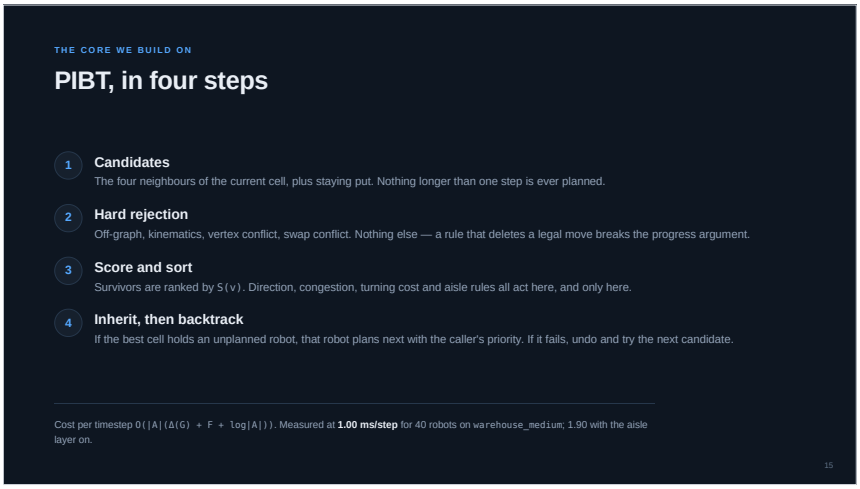
SPEAKER NOTES

That closing question is the thesis of the whole project, so state it as a question and let the results answer it. The short version of the answer, for anyone who asks now: on corridor maps yes, on open grids no.



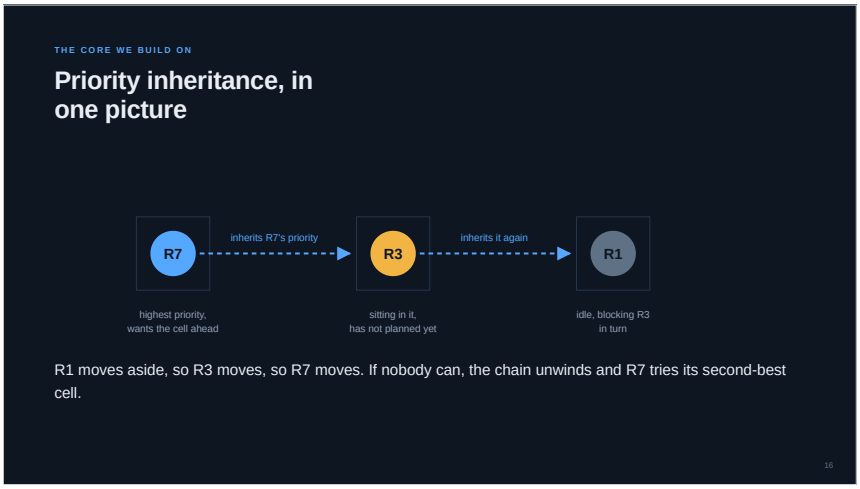
SPEAKER NOTES

The line that matters most is the last one on the right. Because every plan is provisional, a mechanism that is expensive to compute has to justify itself every single timestep — which is the cost argument behind the 56–356× numbers later.



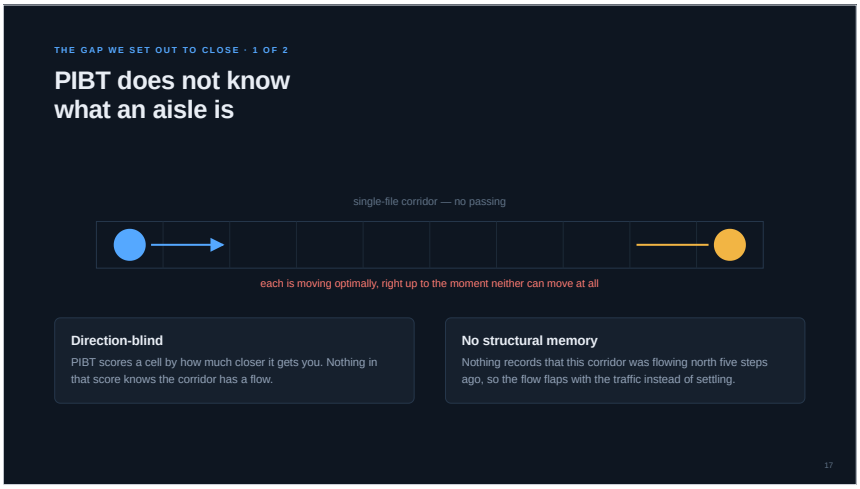
SPEAKER NOTES

Step 2 is the one to emphasise, because the central defect found in this review was that our aisle layer had quietly added rules to it.



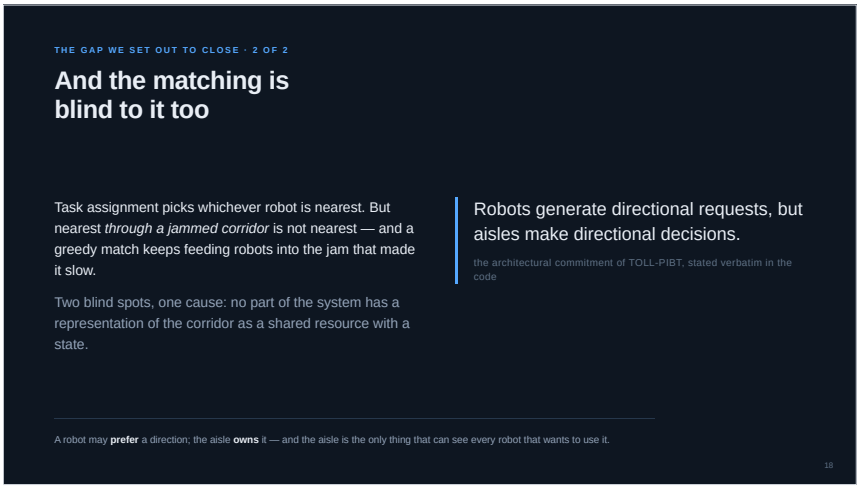
SPEAKER NOTES

This is the mechanism the whole deck turns on. The chain only works because R1 has *somewhere to go*. Delete one of R1's legal moves — which is exactly what a hard one-way rule does — and the chain dead-ends, R7 never moves, and the aisle layer that deleted the move gets blamed for the congestion it caused.



SPEAKER NOTES

The picture does the work here. Both robots are behaving correctly by their own objective, which is what makes this a coordination failure rather than a bug.



SPEAKER NOTES

The quote is the design commitment, and every mechanism in Act 4 follows from it. If someone challenges the whole approach, this is the sentence to defend.

SPEAKER NOTES

The invariant has two halves and people usually only hear the first. No high-level formula can produce a collision — and none can remove a legal move either. The second half is the one the code got wrong.

SPEAKER NOTES

Do not read the formula out. Point at α and ζ and say the ordering matters more than the values, then go to the next slide, which is about exactly that.

λ being both the cheapest term and the most effective mechanism is the single most surprising result in the project. It is worth a pause.

HOW, PART 1

The ordering is the design: α > ζ > β > γ > λ

A whole step of progress beats everything. Driving the wrong way down an aisle beats every other preference. A single turn is the cheapest thing you can be asked to pay.

ζ < α, and ζ is a price

The aisle terms are penalties, not rejections. A robot drives the wrong way when that is the only way to make progress — and pays for it.

μ·C must be normalised

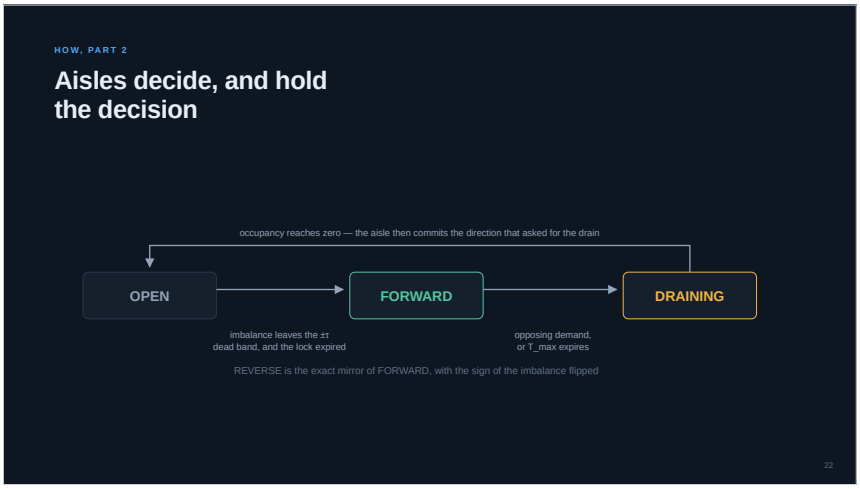
Unnormalised, the congestion mixture is a robot *count* and μ·C reaches the scale of α·Δ — measured mean 3.40, p90 5.75, max 9.60. Congestion then outranks the terms it is meant to modulate.

Both conditions were false in the code until this review. The ordering the score claimed was not the ordering it had.

21

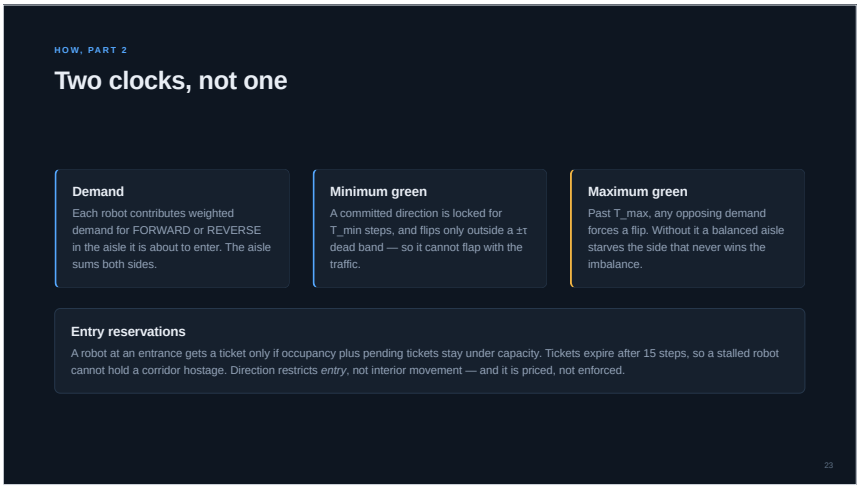
SPEAKER NOTES

This is the technical heart of the review. Two independent bugs, both of which made the same class of mistake: a mechanism meant to modulate was given enough weight to overrule.



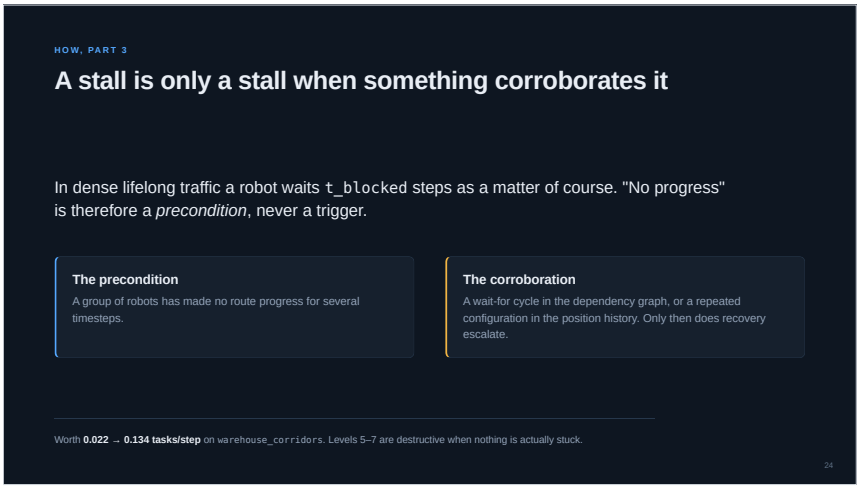
SPEAKER NOTES

The drain state is the non-obvious one. You cannot reverse a corridor that still has robots in it, so a flip is always a two-phase commit: empty it, then turn it.



SPEAKER NOTES

Hysteresis alone is only half a traffic signal: it bounds how soon a direction may change, and nothing bounds how long it may persist. Pickups on one side of the map and deliveries on the other keeps demand permanently balanced, so without the maximum green an aisle holds one direction forever.

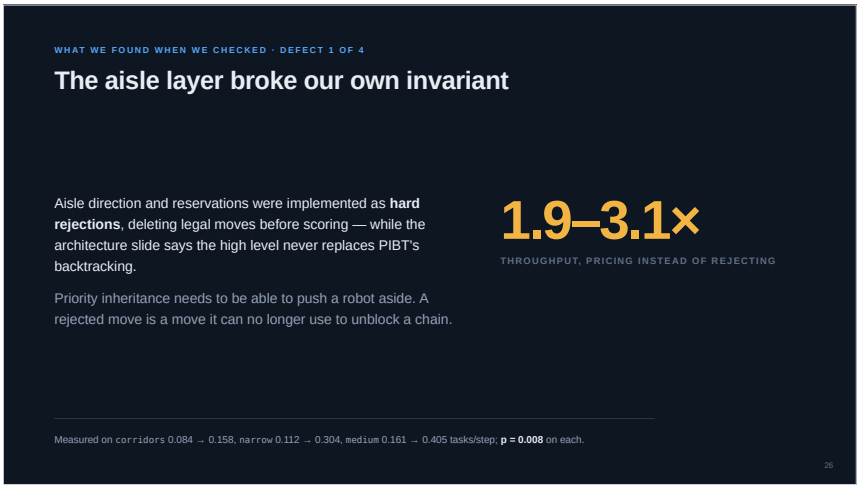


SPEAKER NOTES

This effect was invisible before the other fixes landed, because the hard aisle constraints were manufacturing real deadlocks for recovery to rescue. Remove the cause and the false positives show through — a good example of why fixing bugs in order matters.

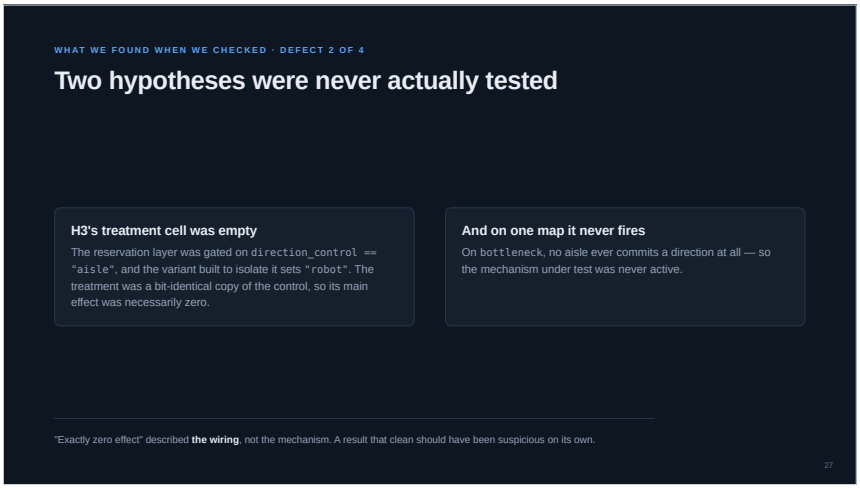
SPEAKER NOTES

The ladder is ordered by how much it gives up. Level 1 costs nothing but CPU; level 7 abandons the entire contribution of this project locally. That ordering is deliberate — you spend the cheap remedies first.



SPEAKER NOTES

This is the largest single effect in the repository, and it came from deleting code rather than adding any. Say that plainly — it is the most useful lesson in the deck.

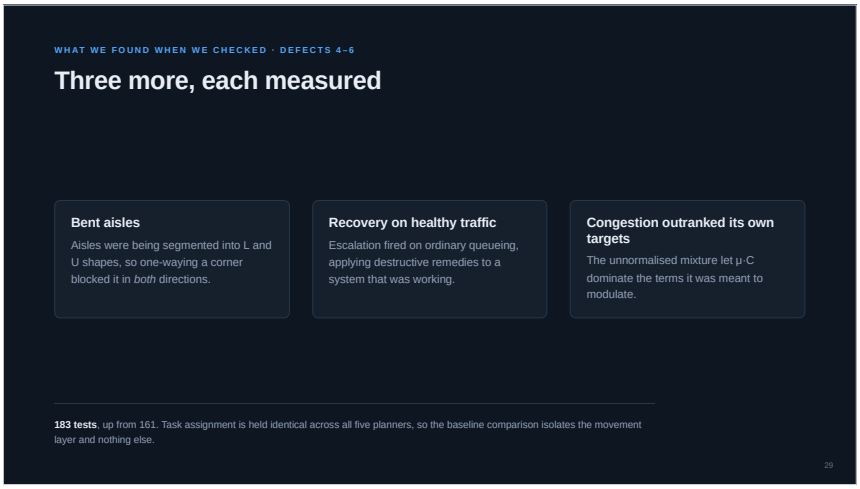


SPEAKER NOTES

The methodological point: an effect size of exactly zero, to full precision, is nearly always a bug. Real mechanisms are noisy even when they do nothing.

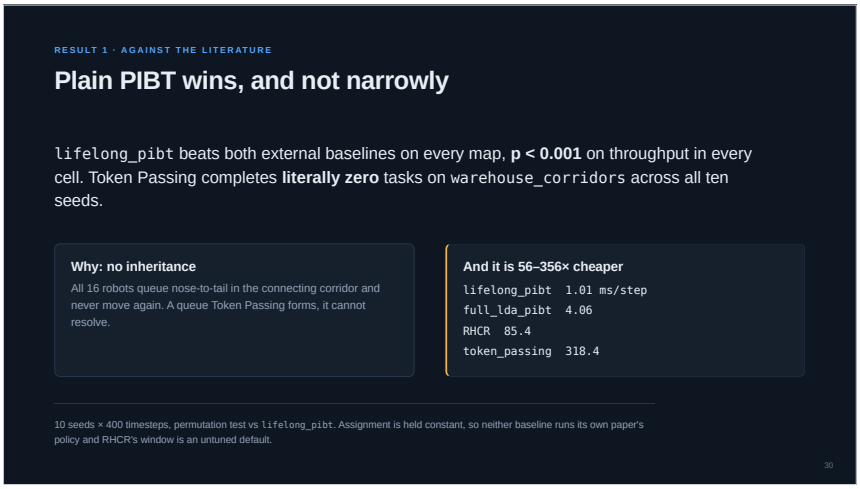
SPEAKER NOTES

Note the interaction with defect 1: the maximum green was decisive while direction was a hard constraint, because there was no other way out. Once counterflow is merely expensive, a robot can buy its way out, so the two fixes are partly redundant. It stays because it makes the layer correct, not merely survivable.



SPEAKER NOTES

Each of these has a regression test now. The test count going up by twenty-two is the durable part of this review; the throughput numbers will move again, the tests will keep the mechanisms honest.



SPEAKER NOTES

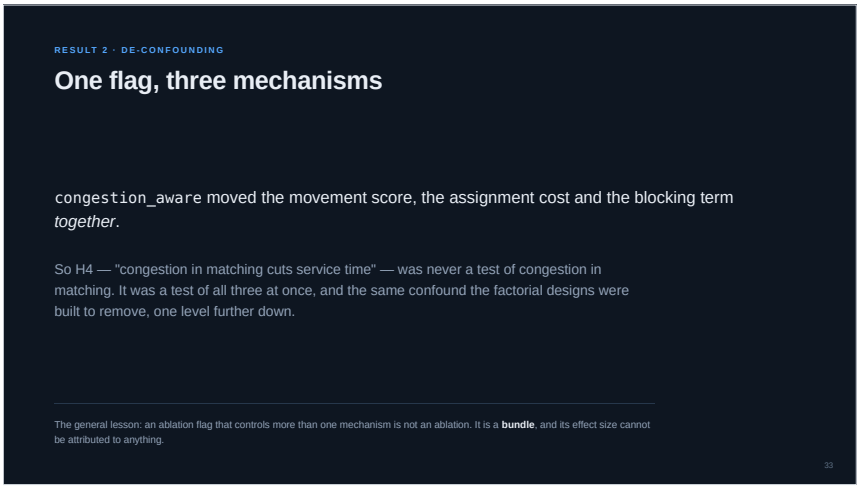
Do not oversell this. The caveat in the footer is real: we hold assignment constant to isolate the movement layer, which means neither baseline is running the full system its authors published. It is the right experiment for our question and the wrong one for "is RHCR good".

SPEAKER NOTES

This slide is the answer to the thesis question posed on slide 13. Say it as a finding, not an apology: a mechanism with a map-dependent payoff is a useful result, provided you can say which maps.

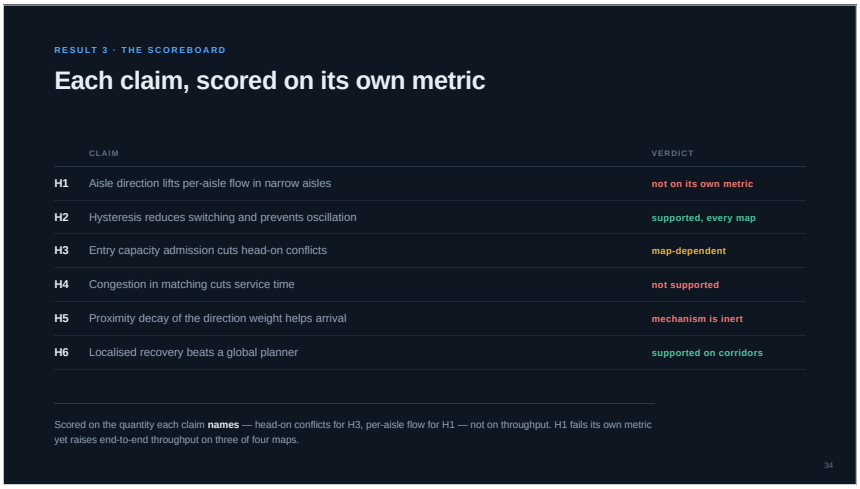
SPEAKER NOTES

The interaction is the interesting part: both together is worse than admission alone. That suggests entry admission is not merely useless but actively fights the direction signal, which is the first item on the future-work slide.



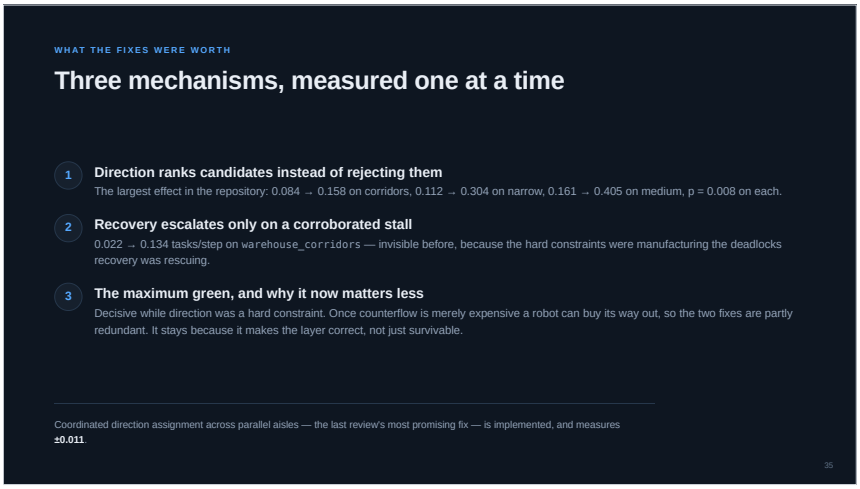
SPEAKER NOTES

Keep this slide sparse. It is a methodology point and it lands better as one sentence than as a table.



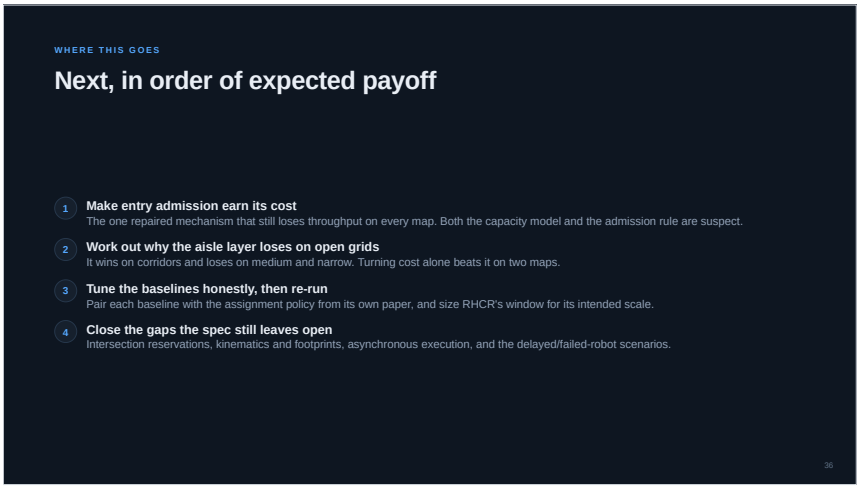
SPEAKER NOTES

The footer is the point of the whole slide. Scoring every hypothesis on throughput would have marked H1 as a success and told us nothing about why. Scoring it on per-aisle flow tells us the mechanism does not do what we said it does, even though the system is better with it.



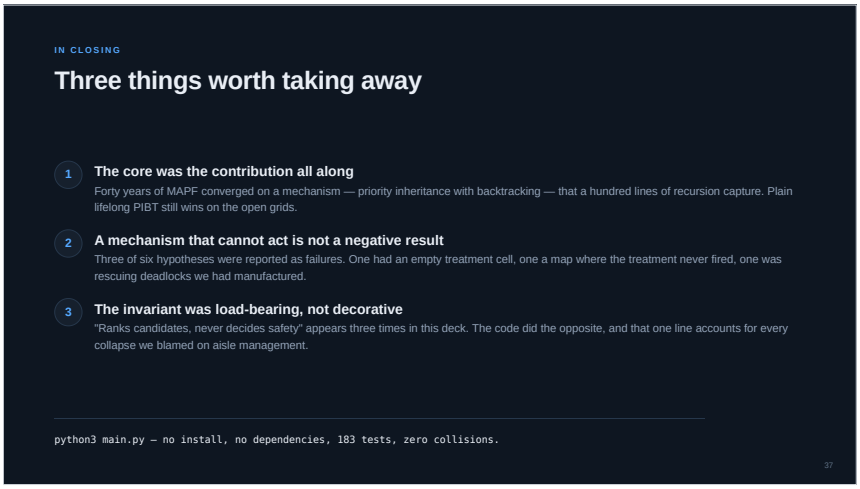
SPEAKER NOTES

The footer deserves a sentence out loud: the fix we expected most from did nothing measurable, and the fix that mattered most was removing code. It is worth being explicit that our predictions were poor.



SPEAKER NOTES

Item 3 is the one a reviewer will press on, and it should be conceded early: our baseline numbers answer "which movement layer is better", not "which published system is better".



SPEAKER NOTES

End on the second point if there is time for only one. The most transferable lesson from this project is that a failed hypothesis deserves the same scrutiny as a successful one, because the most common cause of a clean null result is a treatment that never ran.